

LECTURE 11

QUIZ 1 - 15 Minutes

Design Thinking Vs System Thinking

Design Thinking and Systems Thinking are complementary approaches. They can be used independently, but they are much more powerful when used together, especially for complex products, services, organisations, and social problems.

Design Thinking Vs System Thinking

Human Lens

- Who?
- What do they need?
- What do they experience?
- What matters to them?

System Lens

- What is happening?
- Why does it keep happening?
- Who and what are connected?
- What structures produce the behaviour?
- What might happen next?

Design Thinking Vs System Thinking

	Design Thinking	Systems Thinking
Primary focus	People and their needs	The whole system and relationships
Main question	<i>Are we solving the right problem for people?</i>	<i>Why does this problem exist and keep recurring?</i>
Perspective	Human-centred	System-centred / holistic
Looks at	Users, experiences, needs, behaviours	Relationships, structures, feedback, delays
Typical outcome	A desirable solution	Understanding of system behaviour and leverage points
Risk if used alone	May solve the user's immediate problem but ignore wider consequences	May understand the system well but struggle to create desirable human-centred solutions

When Should You Use Design Thinking?

Design Thinking is particularly useful when the problem involves people, uncertainty, unmet needs, experiences, or innovation.

Typical situations : Designing a new product

For example: Students struggle to manage their academic workload.

When Should You Use Design Thinking?

Design Thinking helps ask:

- Who are the students?
- What are they struggling with?
- What do they currently do?
- What frustrates them?
- What do they actually need?

When Should You Use Design Thinking?

Good use cases

- Designing a mobile app
- Improving a website
- Designing a physical product
- Improving customer experience
- Improving hospital patient experience
- Designing educational experiences
- Improving student services
- Developing new services
- Innovation and product development

When Should You Use Systems Thinking?

Systems Thinking becomes particularly important when the problem is:

- Complex
- Recurring
- Involving multiple stakeholders
- Producing unintended consequences
- Influenced by feedback loops
- Changing over time

Example: Traffic congestion

When Should You Use System Thinking?

A Design Thinking approach may ask:

How can we make commuting easier for citizens?

But Systems Thinking asks:

- Why does congestion keep returning?
- What happens when we build more roads?
- Does increased road capacity encourage more cars?
- What role does public transport play?
- What incentives influence people's choices?
- Are there delays?
- Are there reinforcing loops?

When Should You Use System Thinking?

Good use cases

- University Student Attendance
 - **Problem:** Students are not attending classes regularly.
- Food Delivery Ecosystem
 - **Problem:** How can delivery become faster?
- Hostel Wi-Fi and Internet Bandwidth
 - **Problem:** The internet becomes slow every evening.
- Product User Complaints 
 - **Problem:** Users keep complaining despite frequent software updates.
- Hospital Emergency Department 
 - **Problem:** Patients experience long waiting times.

When Should You Use System Thinking?

Good use cases

- Project Delays in Software Teams
 - Problem: Every project misses its deadline.
- Campus Waste Management
 - Problem: The campus generates excessive waste.
- Employee Burnout in an Organisation
 - Problem: Employees are increasingly stressed and leaving.
- Water Shortage
 - Problem: A city experiences water shortages.
- Education and Examination System
 - Problem: Students focus on marks rather than learning.

When Should You Use System Thinking?

A Good Rule of Thumb for Students

Use Systems Thinking when a problem involves multiple interconnected elements, multiple stakeholders, feedback loops, delays, recurring behaviour, or unintended consequences.

This makes Systems Thinking especially valuable for complex problems where fixing one part can affect many other parts of the system.

Design Thinking without Systems Thinking can produce:

"A beautiful solution to the wrong systemic problem."

For example: Build a better flyover.

The flyover may be:

- Well designed
- User-friendly
- Efficient
- Aesthetically excellent

But it may still increase:

- Car dependency
- Induced demand
- Urban sprawl

Systems Thinking without Design Thinking can also fail:

Suppose you perfectly understand:

- Feedback loops
- System structures
- Incentives
- Delays
- Stakeholders

But then you design a solution without understanding users.

You may produce: A technically perfect system that nobody wants to use.

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You may produce: A technically perfect system that nobody wants to use.

Systems Thinking without Design Thinking can also fail:

For example:

A university develops a highly efficient

"Automated academic advising system."

But students find it Confusing, Impersonal and Difficult to access

What we produced? Systemically efficient but Humanly poor.

This is why Design Thinking matters.

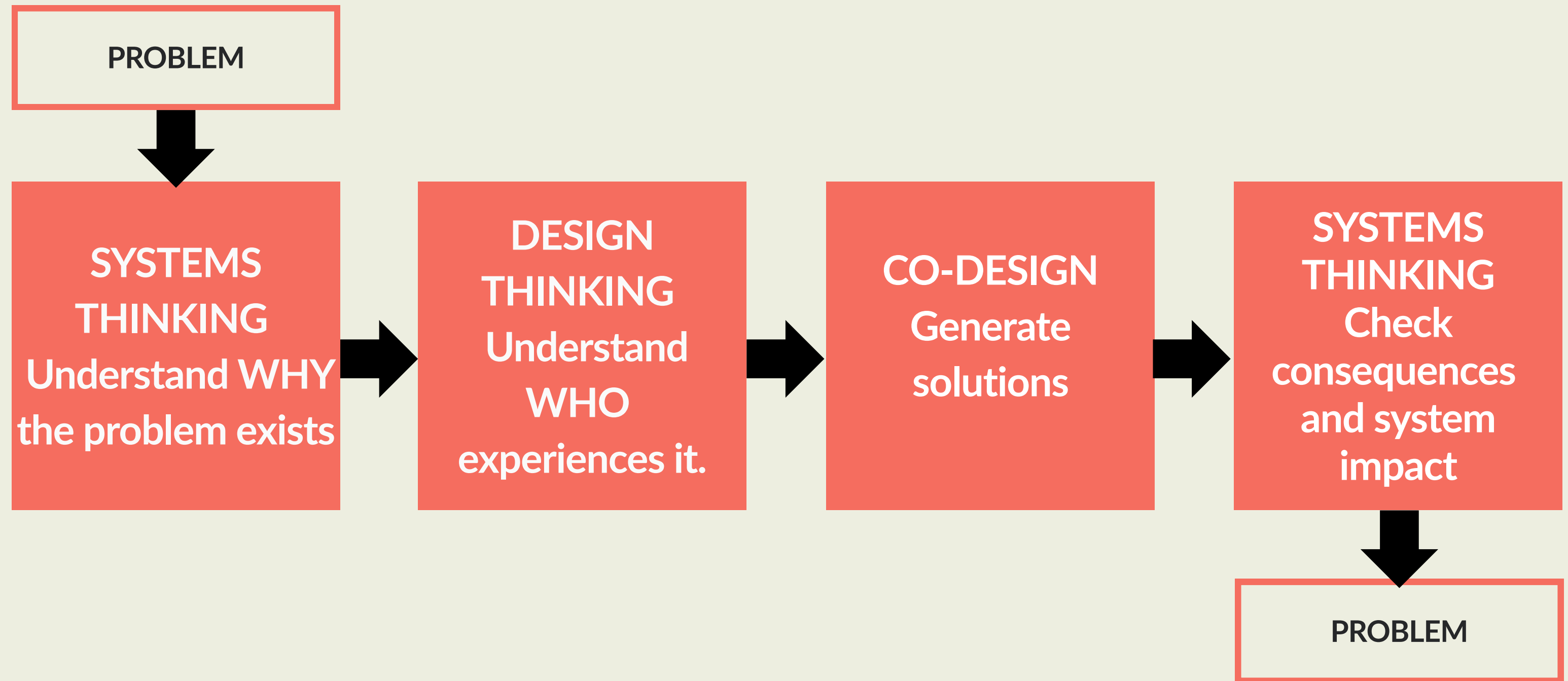
Design Thinking + System Thinking

Design Thinking *helps us design the right solution for people.*

Systems Thinking *helps us understand whether that solution addresses the deeper problem and how it will affect the larger system.*

They should ideally work together.

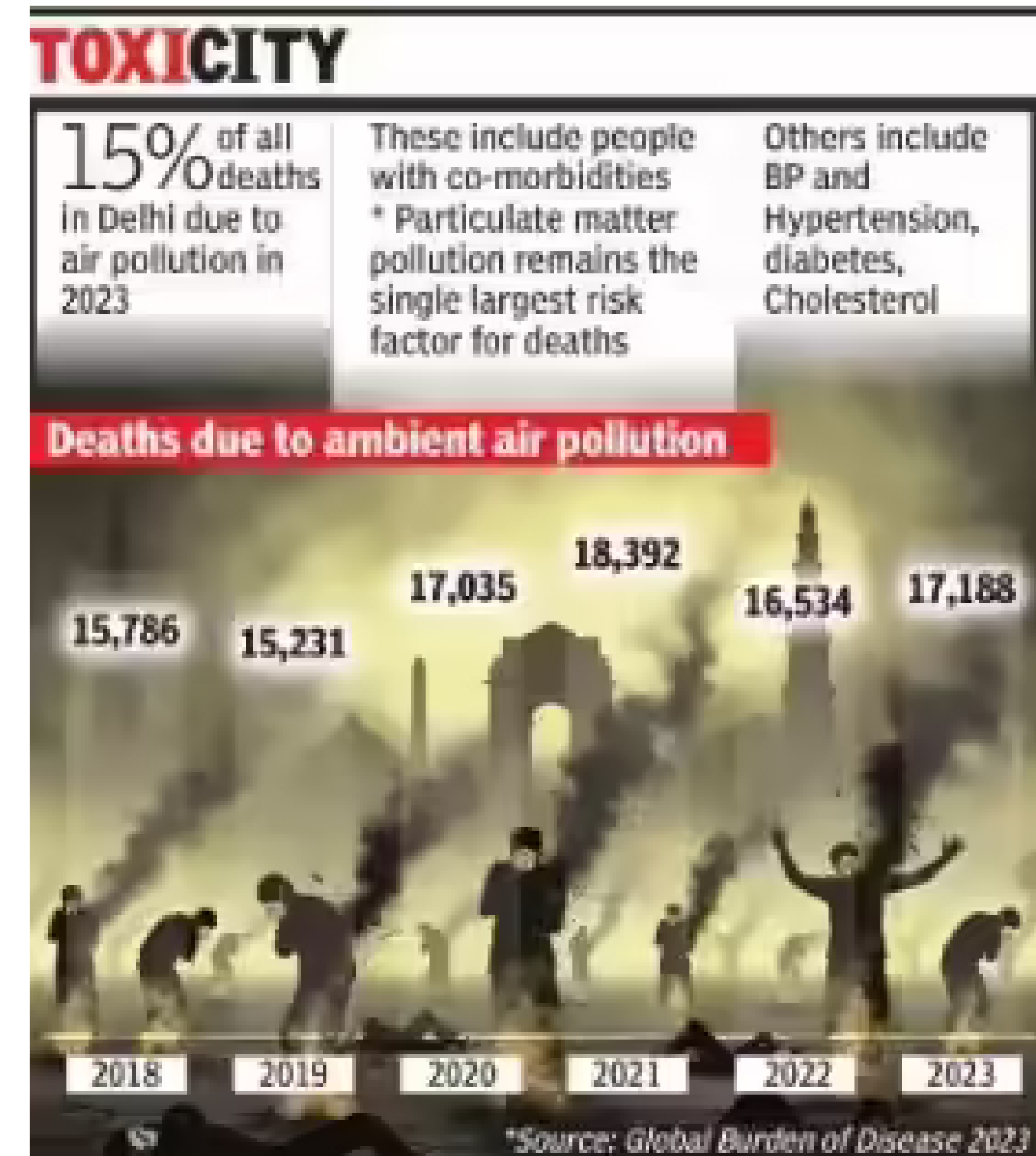
The Ideal Relationship



LECTURE 12

- In India overall, air pollution is estimated to cause over 2 million premature deaths each year, making it a leading environmental risk factor nationwide.
- Approximately 17,188 deaths in Delhi linked to Air Pollution in 2023

Metric	Typical Value
WHO annual PM2.5 guideline	5 $\mu\text{g}/\text{m}^3$
Delhi winter average PM2.5	$\sim 178 \mu\text{g}/\text{m}^3$
National PM2.5 standard (India)	40 $\mu\text{g}/\text{m}^3$
Annual average AQI (2025)	~ 201
Annual average AQI (2026 est.)	~ 280
PM10 annual average	$\sim 200 \mu\text{g}/\text{m}^3$



How do we **resolve** this?

Who should take responsibility?

Activity

- Form a group of 3 members
- and...

Applying Systems Thinking to the Delhi Air Pollution Challenge

- STEP 1 Define the Problem
- STEP 2 Define System Boundary
- STEP 3 Identify Stakeholders
- STEP 4 Identify Events
- STEP 5 Identify Patterns Over Time
- STEP 6 Identify System Structures
- STEP 7 Identify Mental Models
- STEP 8 Identify Variables
- STEP 9 Map Causal Relationships
- STEP 10 Identify Feedback Loops
- STEP 11 Identify Delays
- STEP 12 Identify System Archetypes
- STEP 13 Find Leverage Points
- STEP 14 Develop Multi-Level Interventions
- STEP 15 Monitor System Behaviour Over Time

LECTURE 13 -7 Sep

LECTURE 14 - 10 Sep

JEE coaching phenomenon

Conventional Problem Statement

The Indian JEE preparation ecosystem has become increasingly dependent on intensive coaching, creating high academic pressure, long preparation hours, significant financial and emotional costs, and an unequal educational environment, while encouraging students to optimise for examination performance rather than deep learning, creativity and holistic development.

This is a primarily a problem statement about symptoms.

JEE coaching phenomenon

Systems Problem Statement

The JEE preparation system has evolved into a self-reinforcing ecosystem in which high-stakes selection, limited perceived pathways to prestigious engineering institutions, parental aspirations, competitive student behaviour, coaching-market incentives, school-coaching dynamics and social perceptions of success collectively increase dependence on coaching. Each actor's individual rational decisions unintentionally reinforce the overall system, creating escalating competition, academic pressure, inequality and an increasingly narrow definition of educational success.

JEE coaching phenomenon

The key Systems Thinking move is to avoid asking:

“How do we fix coaching?”

and instead ask:

“What is the system structure that makes intensive JEE coaching rational, self-reinforcing and difficult to escape?”

First Principle: Don't Blame the Coaching Institutes

This is critical, Instead ask:

Why do students go to coaching?

Because parents believe coaching improves their chances.

Why do parents believe that?

Because the examination is highly competitive.

Why is the examination competitive?

Because there are many aspirants competing for relatively limited seats in highly prestigious institutions.

Why are those institutions so attractive?

Because of perceived career opportunities, social status and economic mobility.

Why does this matter so much?

Because engineering education is seen by many families as a relatively secure pathway to upward mobility.

JEE coaching phenomenon

So we have a system:

Social aspiration → JEE competition → Coaching → Higher preparation intensity → Higher competitive benchmark → Greater perceived need for coaching → More coaching

That is a reinforcing feedback loop.

Define the System Boundary

What is inside our system? For example:

- JEE aspirants
- Parents
- Schools
- Coaching institutes
- Online coaching platforms
- Teachers
- JEE examination system
- IITs/NITs and other engineering institutions
- Students' peer networks
- Test-preparation technology
- Educational media
- Rankings and result publicity
- Scholarships
- Career opportunities
- Social perceptions of success

Define the System Boundary

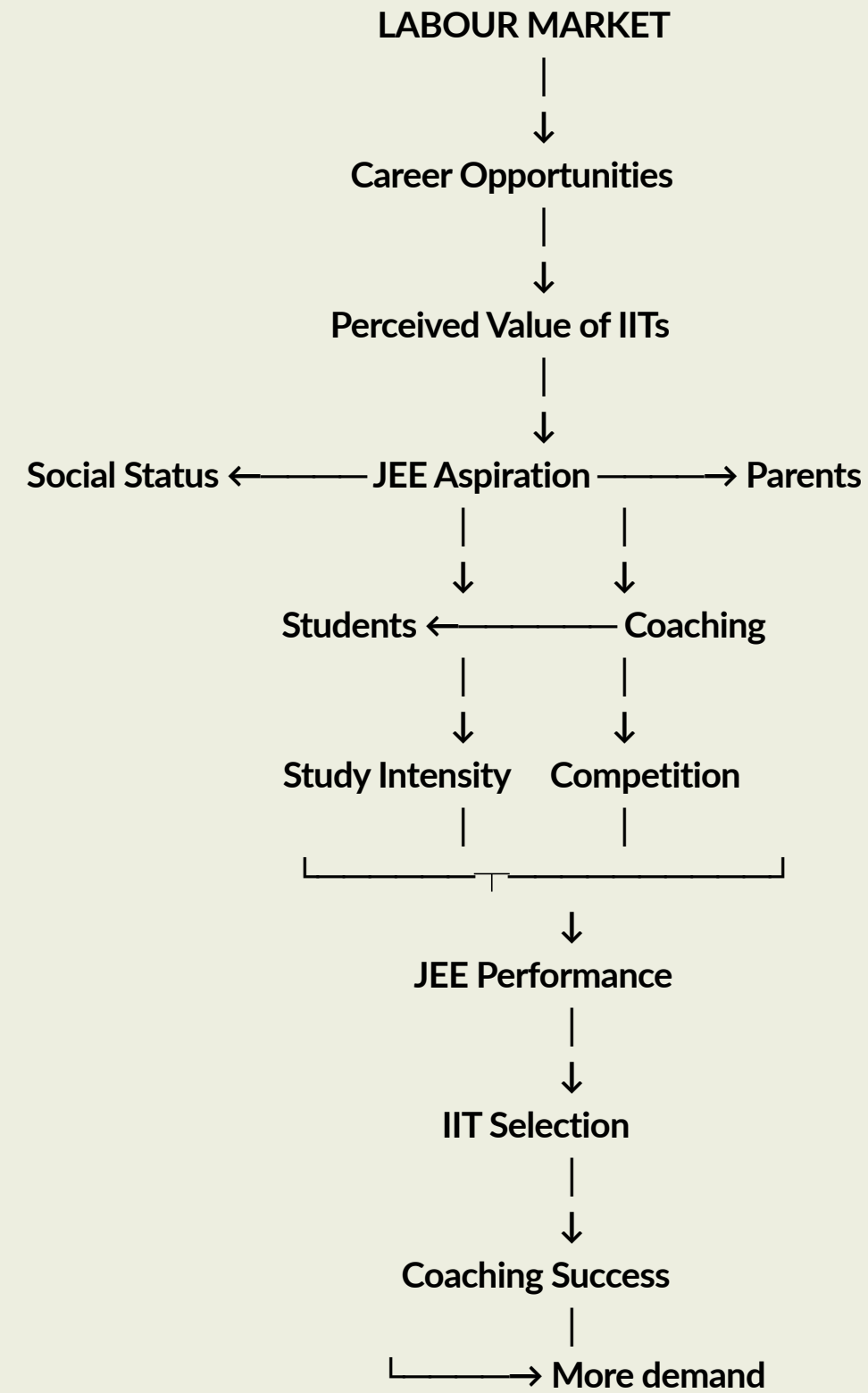
What is outside but influences the system?

- Labour market
- Economic inequality
- Family income
- Social mobility
- Higher-education policy
- Government education policy
- Corporate recruitment
- Broader cultural attitudes toward success

This immediately shows that:

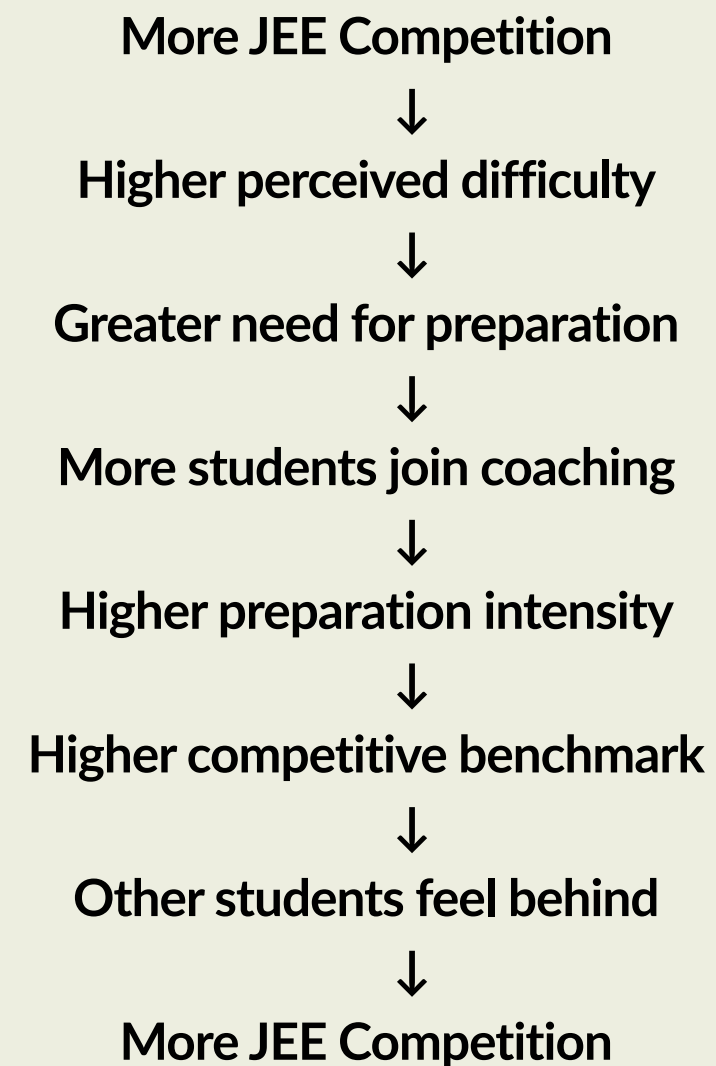
JEE coaching is not an isolated education problem. It is an ecosystem problem.

Map the Actors



identify causal relationships and feedback loops

The Competition Escalation Loop - R1



This creates a classic: Reinforcing Loop – R1. The system continuously escalates.

The Coaching Dependence Loop

Consider a student who initially thinks:

“I can prepare through school and self-study.”

Then the student discovers that many peers are attending coaching.

They begin thinking:

“Maybe I am falling behind.”

So:

More peers in coaching → greater perceived disadvantage → higher coaching demand → more students in coaching

Coaching Normalisation –R2

More students join coaching



Coaching becomes socially normal



Non-coaching preparation appears risky



More students join coaching

Eventually, coaching becomes not merely an option, but the default expectation.

The Results–Reputation Loop

Coaching institutes advertise successful candidates.

Suppose an institute produces many high-ranking students.

Parents see:

“Students from this institute get into IIT.”

Demand increases.

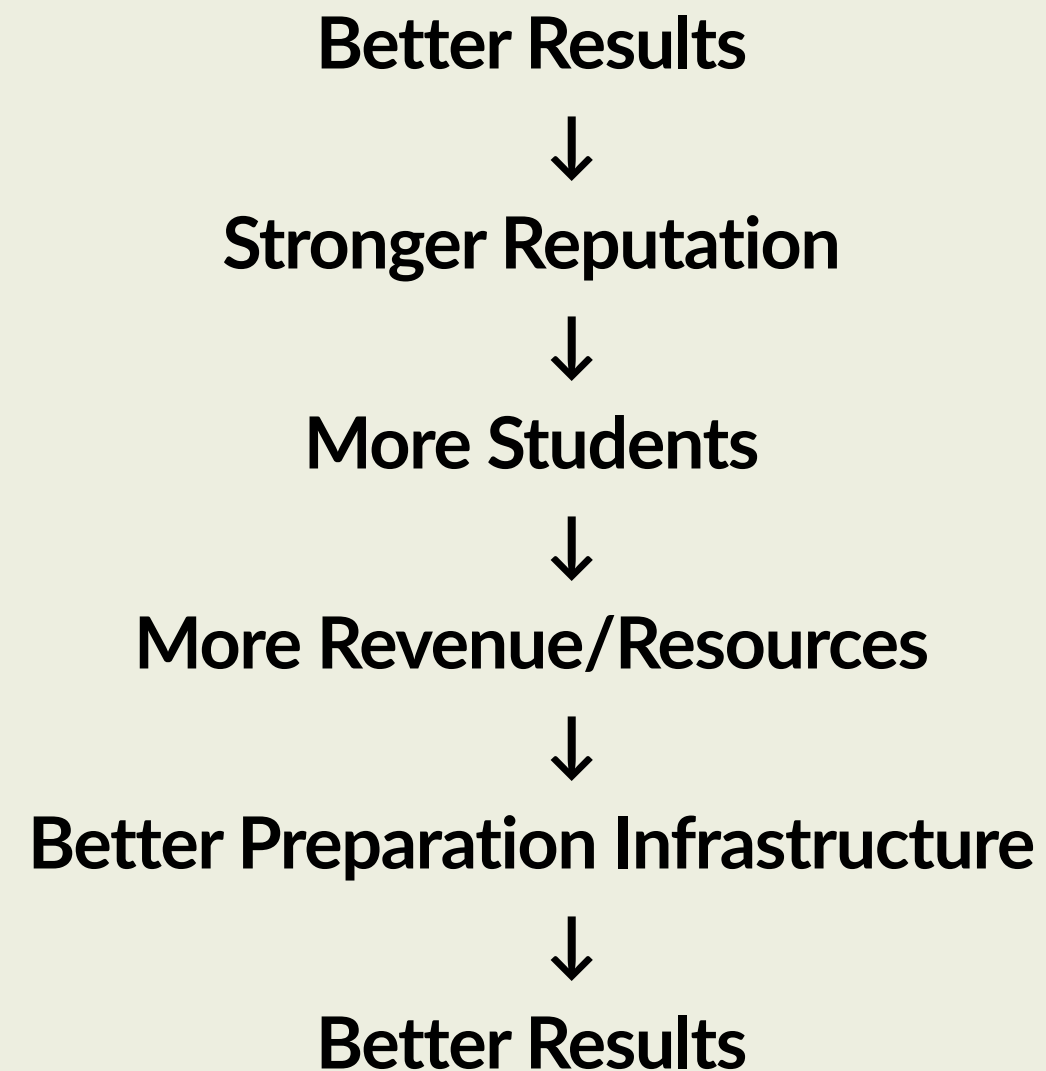
More students join.

The institute gets more resources.

It can hire better teachers, develop better material and run more tests.

Performance improves further.

Coaching Success Loop – R3



Eventually, coaching becomes not merely an option, but the default expectation.

The Selection Effect

Suppose a coaching institute claims:

“We produced 500 IIT selections.”

But who joined the institute? Often, students who were already:

- academically strong,
- highly motivated,
- financially supported,
- selected through internal screening,
- willing to study intensively.

therefore: **Strong students → Coaching → Strong results**

may be interpreted as: **Coaching → Strong students**

This is a potential selection-effect problem.

Distinguish: Correlation from causation.

This is important when evaluating whether more coaching necessarily produces better outcomes.

The School–Coaching Tension

Students have limited time.

If coaching becomes increasingly demanding:

Coaching hours $\uparrow$ $\rightarrow$ Time available for school $\downarrow$

Schools then find students less engaged.

Students perceive school as less relevant to JEE.

Therefore:

School perceived value $\downarrow$ $\rightarrow$ Greater dependence on coaching $\uparrow$

Which further weakens the school–student relationship.

This creates a balancing/declining loop around mainstream schooling.

The Inequality Loop – Opportunity Accumulation–R4

High-quality coaching often requires:

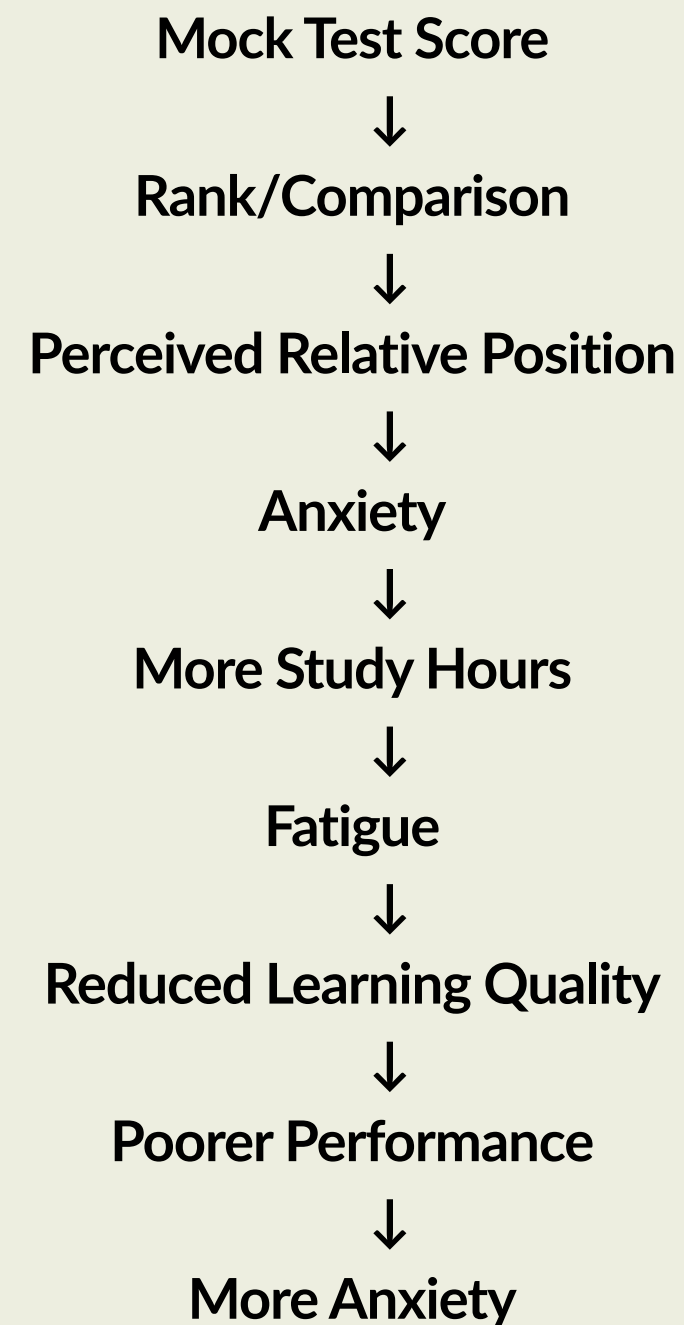
- money,
- time,
- access,
- technology,
- supportive parents,
- relocation or residential facilities.

Students with greater resources can access more preparation.

The important insight is:

The system can amplify pre-existing advantages even when every individual actor is simply trying to maximise their own child's chances.

The Psychological Competition Loop



This can create a balancing loop that becomes dysfunctional.

The student works harder to improve performance, but excessive effort eventually reduces effectiveness.

The Critical Systems Insight

At this point, we can see that the problem isn't:

“Students study too much.”

Nor:

“Parents put too much pressure.”

Nor:

“Coaching institutes exploit competition.”

All of these may be partially true, but the deeper insight is:

Each actor is responding rationally to the incentives and information available to them, while their collective behaviour creates an undesirable system outcome.

Look Beneath the Events

Level 1 — Events

What do we observe?

- Students attending coaching.
- Long study hours.
- Stress.
- Multiple mock tests.
- Parents spending large amounts.
- School attendance declining.
- Coaching advertisements.
- Intense competition.

Look Beneath the Events

Level 2 — Patterns

What trends do we see?

- Coaching participation increasing.
- Competition intensifying.
- Preparation beginning earlier.
- More testing.
- Increasing study hours.
- Growing dependence on coaching.

Look Beneath the Events

Look Beneath the Events

Level 3 — System Structure - *What creates these patterns?*

- High-stakes examination.
- Limited prestigious seats.
- Social value attached to IITs.
- Information asymmetry.
- Competitive peer behaviour.
- Coaching business incentives.
- Parent risk aversion.
- Ranking mechanisms.
- Labour-market signalling.

Look Beneath the Events

Look Beneath the Events

Level 4 — Mental Models - What beliefs sustain the system?

Perhaps:

- *“IIT is the best pathway to success.”*
- *“Without coaching, I cannot compete.”*
- *“More study hours mean better results.”*
- *“A higher JEE rank means a better future.”*
- *“If everyone else is doing coaching, I cannot afford not to.”*
- *“Failure in JEE is failure in life.”*

These mental models may be among the deepest leverage points.

LECTURE 15 - 17 Sep

Leverage in Systems Thinking

Leverage is the idea that a small, well-chosen change in a system can produce a disproportionately large and lasting impact.

In simple terms:

Leverage means finding the right place to intervene in a system—not simply applying more effort.

A system may have hundreds of visible problems. We can spend enormous resources fixing symptoms, yet little changes. Systems Thinking asks:

“Where can we make a change that fundamentally influences the behaviour of the whole system?”

That place is called a leverage point.

Leverage in Systems Thinking

The Basic Idea

Imagine you are trying to improve a system. You have two choices:

Option A: Apply more effort

- Add more people
- Spend more money
- Create more processes
- Fix problems repeatedly

Option B: Find leverage

- Change an important relationship
- Improve the flow of information
- Modify incentives
- Change a rule
- Redefine the goal

The second approach may require less effort but produce greater systemic change.

A simple principle: Not all interventions are equally powerful. Some changes affect only a small part of the system.

Leverage in Systems Thinking

A Simple Example: Traffic Congestion

Suppose a city has severe traffic congestion.

A common reaction might be:

Low-leverage intervention

Build more roads.

This may temporarily reduce congestion.

or their wider system-level effects.

This may temporarily reduce congestion, but over time:

Roads become easier to use.

More people choose cars.

More cars enter the roads.

Congestion returns.

The intervention addressed a symptom.

Leverage in Systems Thinking

Higher-leverage intervention

Instead, the city might examine:

- Why do people prefer private cars?
- Is public transport reliable?
- Are workplaces concentrated in one area?
- Are working hours synchronized?
- Are parking policies encouraging car use?

Potential interventions:

- Improve public transport reliability.
- Introduce flexible work hours.
- Change parking incentives.
- Improve walking and cycling infrastructure.

These interventions affect the structure that generates congestion.

That is leverage.

Leverage in Systems Thinking

Different Levels of Leverage

We can think of leverage as existing at several levels.

Level 1: Parameters

These are numerical adjustments.

Examples:

Leverage in Systems Thinking

4. Different Levels of Leverage

We can think of leverage as existing at several levels.

Level 1: Parameters

These are numerical adjustments.

Examples:

Increasing budget

Adding staff

Increasing capacity

Reducing cost

Changing deadlines

Example

A hospital has long waiting times.

The hospital adds:

10 more staff members.

This may help.

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But the underlying problem may remain.

LECTURE 15 - 17 Sep

Find the Leverage Points

This is where Systems Thinking becomes useful for designing solutions.

We shouldn't ask:

“How do we eliminate coaching?”

Instead:

“Where can we intervene so that the system no longer requires ever-increasing coaching intensity?”

Six Leverage Points

Leverage Point 1: Change the Information Structure

Today students and parents often have limited information about:

- Alternative career pathways
- Engineering colleges beyond a small set of prestigious institutions
- Actual relationship between coaching intensity and outcomes
- Different learning approaches
- Long-term career outcomes

Six Leverage Points

Leverage Point 1: Change the Information Structure - Intervention

Create a transparent engineering education and career information ecosystem.

Students could see:

- Multiple engineering pathways
- College outcomes
- Branch-level career trajectories
- Skills required
- Alumni outcomes
- Alternative entrance routes
- Cost-benefit information

The objective is not to tell students: “Don't pursue IIT.”, It is to allow them to make better-informed decisions.

Six Leverage Points

Leverage Point 2: Change the Goal

The current system can implicitly optimise for:

Rank → Selection → Institution

We could broaden the educational goal to:

Learning → Capability → Problem-solving → Career readiness

This changes, what the system values. The question becomes:

Can we design an entrance system that selects for genuine capability rather than rewarding increasingly specialised test preparation?

This is a much deeper intervention.

Six Leverage Points

Leverage Point 3: Change Feedback

Currently, one powerful feedback signal is:

Mock-test score → Rank → Anxiety → More preparation

What if we introduce richer feedback?

Instead of simply: Your score = 187/300 - students receive:

Then the feedback loop changes from:

Score → Competition

to:

Evidence → Understanding → Improvement → Capability

This could significantly change student behaviour.

- Concept mastery
- Reasoning ability
- Problem-solving patterns
- Misconception analysis
- Learning progress
- Transfer ability
- Areas of strength
- Areas requiring improvement

Six Leverage Points

Leverage Point 4: Change Incentives

If the system rewards only final examination performance, students and institutions naturally optimise for that metric.

What if the broader ecosystem rewarded:

- Deep understanding
- Research exposure
- Problem-solving
- Creativity
- Projects
- Olympiads
- Innovation
- Interdisciplinary capability

Then preparation behaviour would begin to change.

The principle is:

People optimise what the system measures and rewards.

Six Leverage Points

Leverage Point 5: Redesign the School–Preparation Relationship

Instead of School vs Coaching

we could create:

School + Open Learning Ecosystem

Schools provide:

- Conceptual foundations
- Mentorship
- Collaborative learning
- Projects
- Well-being support

Six Leverage Points

Leverage Point 5: Redesign the School–Preparation Relationship

An open national digital ecosystem provides:

- High-quality lectures
- Practice problems
- Adaptive assessments
- Doubt support
- Simulations
- Learning analytics

Students could then choose their preparation intensity according to their goals.

This could reduce the need for expensive proprietary coaching.

Six Leverage Points

Leverage Point 6: Change the Mental Model

This may be the deepest intervention.

We need to challenge the equation:

JEE Rank = Intelligence = Future Success = Personal Worth

A healthier mental model is:

JEE is one selection mechanism for one set of educational pathways—not a measure of human potential or future success.

This doesn't require reducing the aspiration for excellence.

It means decoupling excellence from a single examination outcome.

But Every Intervention Can Create a New Problem

Suppose we reduce the importance of one examination. What might happen?

Coaching institutes may shift to:

- Olympiads
- Interviews
- Portfolio preparation
- School marks
- New entrance examinations

In other words:

The coaching industry may adapt.

Therefore, simply changing the examination may not solve the system.

This is why we need to anticipate second-order effects.

Design the Future System Rather Than "Destroy Coaching"

A better design objective would be:

Create a high-quality, equitable JEE preparation ecosystem in which coaching is an option rather than a perceived necessity.

That is a much more achievable systems goal.

- We don't necessarily eliminate coaching.
- We change the conditions that create excessive dependence on it.

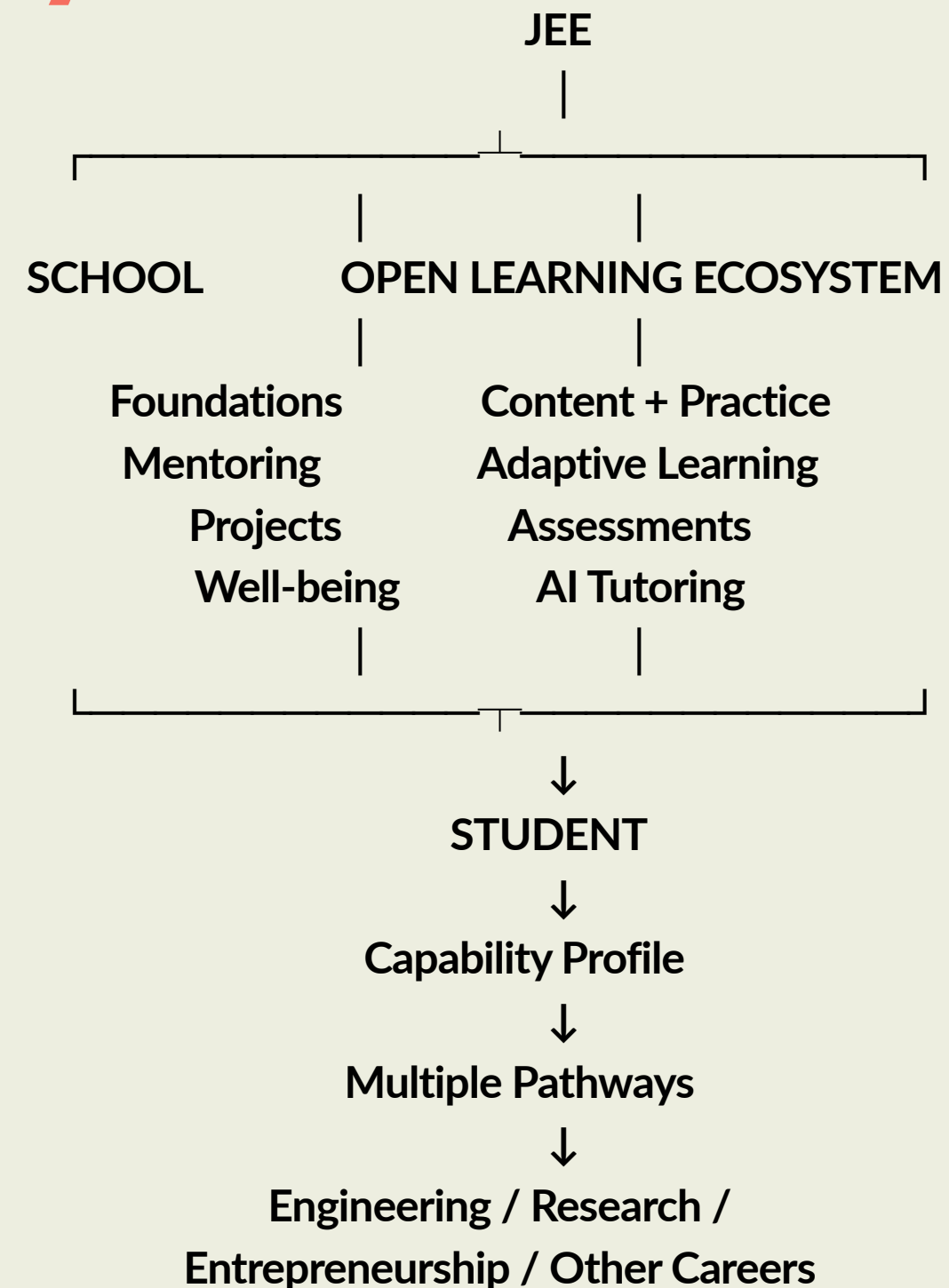
A Proposed System-Level Intervention

Open JEE Learning Ecosystem

Instead of a fragmented:

Student → Coaching Institute → JEE

A Proposed System-Level Intervention



The key design principle is:

Don't compete with coaching at the coaching layer. Change the architecture of the ecosystem.

Test the Intervention Using Systems Thinking

Before implementing nationally, we could create different scenarios.

Scenario A – Current System

High competition + private coaching.

Scenario B

High-quality free digital preparation.

Scenario C

Reduced dependence on single high-stakes testing.

Scenario D

Multiple assessment dimensions.

Scenario E

School + open digital ecosystem.

Test the Intervention Using Systems Thinking

Examine:

- Coaching participation
- Student study hours
- Family expenditure
- Stress
- Equity
- Learning outcomes
- JEE performance
- School engagement
- Unintended consequences

The objective isn't to predict the future perfectly. It is to ask:

How might the system respond to each intervention?

Test the Intervention Using Systems Thinking

Examine:

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The objective isn't to predict the future perfectly. It is to ask:

How might the system respond to each intervention?

Activity : “Causal Loop Challenge”

Objective

Understand feedback loops.

Task Scenario

“College introduces AI attendance monitoring.”

Activity : “Causal Loop Challenge”

Task - Draw:

- Reinforcing Loop (R)
- Balancing Loop (B)

Activity 3: “Causal Loop Challenge”

Example Questions

- Does attendance increase?
- Does student stress increase?
- Does creativity decrease?
- Does admin workload change?

Outcome

Students discover unintended consequences.